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PI Name:	Kubota, Takuo M.D., Ph.D.		
Project Title:	Interaction Between IGF-1 and Integrin Signaling in the Response of Bone to Mechanical Load		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
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PI Email:	tkubota0109@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	415-221-4810
Organization Name:	University of California, San Francisco		
PI Address 1:	Department of Medicine		
PI Address 2:	4150 Clement St		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121-1545	**Congressional District:**	8
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bikle, Daniel (MENTOR/Veterans Affairs Medical Center - University of California)		
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	POSTDOCTORAL FELLOWSHIP Bone loss that results from the lack of mechanical stimulation in the weightlessness of spaceflight is a significant concern for prolonged space exploration missions. However, the mechanisms of bone loss in microgravity have not been fully understood. Evidence supports a role for insulin-like growth factor type 1 (IGF-1) signaling in mediating skeletal response to mechanical load. Skeletal unloading by hindlimb elevation, a ground-based model for the unloading of spaceflight, causes bone loss due to reduced bone formation. IGF-1 is a potent growth factor in bone and is produced in osteocytes and mature osteoblasts after mechanical load. This IGF-1 may produce new osteoblasts and differentiate them into mature osteoblasts to form new bone. Furthermore, skeletal unloading leads to resistance to the anabolic effect of IGF-1. IGF-1 fails to increase bone formation and osteoblast proliferation in the unloaded bone. This failure to respond		

Task Description:	to IGF-1 is due to a failure of IGF-I to stimulate the phosphorylation of the IGF-1 receptor (IGF-1R), although IGF-1R levels and binding of IGF-1 to its receptor are not changed. Skeletal unloading decreases integrin expression in osteoblasts, and this impairment of expression is restored by reloading the bones or the osteoblasts in parallel with the recovery of IGF-1 responsiveness. Knockdown of beta1 and beta3 integrins blocks the ability of IGF-1 to activate its receptor and inhibits its activation by mechanical loading. We will use axial cyclic loading of the tibia as well as reloading following skeletal unloading in mice lacking IGF-1R or beta1 and beta3 integrins to determine the importance of these molecules in the response of bone to mechanical load. We will also use pulsatile fluid flow, a mechanical loading modality in vitro, to determine the molecular mechanism by which integrin signaling by mechanical load enhances IGF-1 signaling. If this hypothesis proves to be true, efforts to maintain the integrin signaling and IGF-1 signaling will be sought to prevent bone loss associated with skeletal unloading. Key findings: Mice lacking the IGF-1R in mature osteoblasts (KO) were hindlimb-unloaded and reloaded after unloading. Total cross-sectional area (Tt.Ar) in cortical bone increased with reloading in control male mice compared to that after unloading. However, the response of Tt.Ar to reloading was completely blunted in KO mice. Cortical bone area was diminished after reloading in KO mice, but tended to be elevated in control mice. Bone formation, including mineralizing surface, mineral apposition rate and bone formation rate, in periosteal bone remained suppressed in KO mice after reloading unlike the complete recovery of control mice. The response of bone volume fraction (BV/TV) of the trabecular bone to unloading and reloading is similar between KO and control mice. However, the percent decrease in BV/TV with unloading was larger in KO mice compared to control mice. The response of bone formation in trabecular and endosteal cortical bone in KO mice was comparable to control mice. Impact of key findings: The above findings indicate that the IGF-1R in mature osteoblasts plays a critical role in periosteal bone formation of cortical bone, but has only a modest role in trabecular bone and endosteal cortical bone, in the skeletal response to unloading and reloading. In trabecular bone and endosteal surfaces of cortical bone, IGF-1 resistance is likely manifested in the less mature osteoblasts and osteoprogenitors which are responding normally in this mouse model. Proposed research plan: We will test mice deleted for integrin beta1 and beta3 from osteobasts to determine whether the skeletal response to mechanical load and IGF-1 infusion is impaired. We will also knock down or inhibit selected components of the integrin/IGF-1R complex to determine whether this blocks IGF-1R activation and/or the formation of the integrin/IGF-1R complex during pulsatile fluid flow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bone loss is a serious problem in immobilized people due to fractures, spinal injuries and strokes on earth. A reduction in bone mass raises a risk of bone fracture. It takes a long time for those patients to recover bone strength in rehabilitation. The mechanisms of bone gain after weight-bearing activity following long-term bed rest have not been fully elucidated as well as those of bone loss during immobilization. Insulin-like growth factor 1 (IGF-1) signaling plays a vital role in bone remodeling. IGF-1 production is increased in osteoblasts after mechanical load. Skeletal unloading by hindlimb elevation, an animal model for immobilization on earth as well as a ground-based model for the unloading of spaceflight, causes a decrease in bone mass associated with resistance to the anabolic effect of IGF-1. Thus, we postulated that IGF-1 signaling mediates the skeletal response to unloading and reloading. To determine whether the IGF-1 receptor (IGF-1R) in mature osteoblasts has a role in the skeletal response to unloading and reloading, we evaluated mice in which the IGF-1R was deleted with osteocalcin-driven cre recombinase. These mice were subjected to skeletal unloading using hindlimb unloading and reloading after unloading. We found that total cross-sectional area (Tt.Ar) in cortical bone increased with reloading in control male mice compared to that after unloading. However, the response of Tt.Ar to reloading was completely blunted in IGF-1R deficient (KO) mice. Cortical bone area was diminished after reloading in KO mice, but tended to be elevated in control mice. Bone formation, including mineralizing surface, mineral apposition rate and bone formation rate, in periosteal bone remained suppressed in KO mice after reloading unlike the complete recovery of control mice. The response of bone volume fraction (BV/TV) of the trabecular bone to unloading and reloading is similar between KO and control mice. However, the percent decrease in BV/TV with unloading was larger in KO mice compared to control mice. The response of bone formation in trabecular and endosteal cortical bone in KO mice was comparable to control mice. These results indicate that IGF-1 signaling in mature osteoblasts plays a critical role in periosteal bone formation of cortical bone, but has only a modest role in trabecular bone and endosteal surfaces of cortical bone, in the skeletal response to unloading and reloading. These results suggest that IGF-1 signaling is necessary for promoting periosteal bone formation after weight-bearing activity following immobilization on earth and that efforts to maintain and potentiate the IGF-1 signaling will be sought to enhance bone formation following long-term bed rest.
Task Progress:	Bone loss is a serious health risk for prolonged space missions. It takes a long time for astronauts to recover bone strength in normal gravity after the microgravity of space. The mechanisms of bone gain after weight-bearing reloading following space flights have not been fully understood as well as those of bone loss during the microgravity of space. Insulin-like growth factor 1 (IGF-1) signaling plays a vital role in bone remodeling. IGF-1 production is increased in osteocytes and osteoblasts after mechanical load. Skeletal unloading by hindlimb elevation, a ground-based model for the unloading of spaceflight, causes a decrease in bone mass associated with resistance to the anabolic effect of IGF-1. Thus, we postulated that IGF-1 signaling mediates the skeletal response to unloading and reloading. To determine whether the IGF-1 receptor (IGF-1R) in mature osteoblasts and osteocytes has a role in the skeletal response to unloading and reloading, we evaluated mice in which the IGF-1R was deleted with osteocalcin-driven cre recombinase. These mice were subjected to skeletal unloading using hindlimb unloading and reloading after unloading. We found that total cross-sectional area (Tt.Ar) in cortical bone increased with reloading in control male mice compared to that after unloading. However, the response of Tt.Ar to reloading was completely blunted in IGF-1R deficient (KO) mice. Cortical bone area was diminished after reloading in KO mice, but tended to be elevated in control mice. Bone formation, including mineralizing surface, mineral apposition rate and bone formation rate, in periosteal bone remained suppressed in KO mice after reloading unlike the complete recovery of control mice. The response of bone volume fraction (BV/TV) of the trabecular bone to unloading and reloading is similar between KO and control mice. However, the percent decrease in BV/TV with unloading was larger in KO mice compared to control mice. The response of bone formation in trabecular and endosteal cortical bone in KO mice was comparable to control mice. These results indicate that IGF-1 signaling in mature osteoblasts plays a critical role in periosteal bone formation of cortical bone, but has only a modest role in trabecular bone and endosteal surfaces of cortical bone, in the skeletal response to unloading and reloading. These results suggest that IGF-1 signaling is required for accelerating periosteal bone formation after

	returning to normal gravity and that attempts to maintain and promote the IGF-1 signaling will be sought to augment bone formation following long-term space missions.
Bibliography Type:	Description: (Last Updated: 07/29/2020)
Abstracts for Journals and Proceedings	Kubota T, Elalieh HZ, Babey M, Wang Y, Bikle DD. "Impact of Deleting the IGF-1 Receptor in Mature Osteoblasts on the Skeletal Response to Unloading and Reloading." 33rd Annual Meeting of the American Society for Bone and Mineral Research, San Diego, CA, September 16-20, 2011. J Bone Miner Res 2011 Sep;26(Suppl 1):SU0071. , Sep-2011
Abstracts for Journals and Proceedings	Kubota T, Elalieh HZ, Wang Y, Babey M, Bikle DD. "IGF-1 receptor in mature osteoblasts and osteocytes is required for reloading induced bone formation." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Articles in Peer-reviewed Journals	Tahimic CG, Long RK, Kubota T, Sun MY, Elalieh H, Fong C, Menendez AT, Wang Y, Vilardaga JP, Bikle DD. "Regulation of ligand and shear stress-induced insulin-like growth factor 1 (IGF1) signaling by the integrin pathway." J Biol Chem. 2016 Apr 8;291(15):8140-9. https://doi.org/10.1074/jbc.M115.693598 ; PMID: 26865633; PMCID: PMC4825016 , Apr-2016